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Leveraging Large Language Models for Cost Management and Supply Chain Optimization

Maryam Shahini, Carina Yu Wang, Marc A. Roeder, Sanjay Pethe, Sarah W. Coffman, Preston Howard, Bo Hu, Hannah N. Wasserman, and Joshua Arnwine, Conocophillips

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Abstract

In this paper, we introduce a cutting-edge solution to the complex challenge of managing operational costs in unconventional asset development, particularly concerning continuous well drilling, completion, and field maintenance operations within the oil and gas sector. These complicated field operations involve hundreds of service providers and incur vast volumes of invoices every year. Tremendous values are lost due to missed opportunities to improve contracting strategy, optimize material and equipment supplies, and identify cost-prohibitive design elements during drilling, completion, production, and plant maintenance through proper spend categorization.

Leveraging the power of a machine learning solution, Large Language Model (LLM), and an interactive user interface, we automate the challenging task of categorizing millions of invoices from diverse service providers. Techniques including sentence transformation embedding, transfer learning, fine-tuning, and re-training process are employed to enhance model performance and adaptability to diverse invoice types.

Through rigorous model training and iterative refinement facilitated by the user interface, our approach attains an impressive accuracy exceeding 90% across all regions. Results prove that the Large Language Model has a wide application in business optimization during unconventional asset development. This automated algorithm provides real-time insights through spending and significantly reduces the turnaround time. This study also enables direct identification of cost-saving opportunities such as potential reductions in fuel expenses related to drilling and completion activities. The advanced analytics capabilities following the modeling effort allow engineers and analysts in multiple functions to identify cost-saving opportunities through customizing visualization tools in various areas.

This paper presents original methodology to adopt the Artificial Intelligence, i.e., Large Language Model, in the oil and gas industry and demonstrate its values with case histories.

Keywords: Invoice Categorization, Artificial Intelligence, Supply Chain Optimization, Machine Learning, Embedding, Large Language Model, Technological Innovation, Predictive Analytics

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Introduction

The oil and gas industry, particularly in the domain of unconventional asset development, faces unique challenges that require innovative solutions to manage operational costs effectively. These assets, characterized by their complex geological settings and intensive resource extraction techniques, demand continuous drilling, completion, and maintenance operations. Such activities involve coordination among hundreds of service providers, each contributing to a vast stream of invoices and logistical data that must be managed and optimized (Fig.1). Historically, the industry has struggled with managing these data flows efficiently, leading to significant financial losses due to outdated contracting strategies, suboptimal supply chain management, and inefficient operational designs.

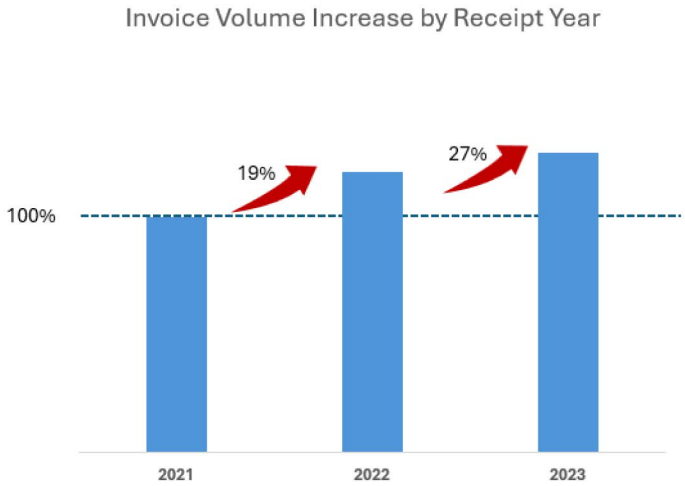

Figure 1—Total invoice increment over the past 3 years for processing and analysis

As the industry continues to mature, there is an increasing awareness of the importance of cutting-edge technologies to address the various challenges associated with unconventional asset development. Traditional methods of data processing and analysis are proving insufficient in handling the vast amounts of data involved in this process, making it difficult to categorize expenses consistently and manage costs strategically. The manual categorization of expenses is not only time-consuming but also prone to errors, emphasizing the need for automated solutions that can efficiently and steadily process large volumes of data. By utilizing advanced technologies such as machine learning and natural language processing, it is possible to streamline cost management processes and improve the accuracy and efficiency of expense categorization, leading to improved profitability and competitiveness in the industry.

Recent advancements in artificial intelligence, particularly in the field of Large Language Models (LLMs), offer promising solutions to these challenges. LLMs, which are trained on a vast amount of text data, have shown remarkable capabilities in understanding and generating human-like text, making them well suited for tasks that require natural language understanding. In the context of the oil and gas industry, LLMs can be employed to automate the processing of invoice data, extracting and categorizing information with high accuracy and speed. This automation not only enhances efficiency but also provides a foundation for deeper analytical insights into spending patterns, enabling more strategic decision making.

Data Processing

In this project, the initial step involves rigorous data processing. We source our data from comprehensive databases, ensuring we have a robust foundation to build upon. However, managing this data is not without its challenges. We contend with large volumes of data, which include varied formats and significant

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inconsistencies that require cleaning and preprocessing. This initial phase is crucial, as it sets the stage for the advanced analytics that allow us to extract meaningful insights from complex datasets.

Data Sources and Characteristics

The primary data source for this study consists of tens of millions of invoice records generated from operational activities in unconventional oil and gas asset development. These records originate from multiple service providers involved in various stages of asset development, including drilling, completion, production, and maintenance. Each invoice contains detailed information such as service descriptions, material descriptions, line spend, quantities, unit prices, and total spend amounts. Due to the collaborative nature of oil and gas projects, these invoices are not only numerous but also varied in format, content, and level of detail.

Challenges

Invoice data often contain inconsistencies and errors due to manual entry, varying data entry standards among providers, and differences in billing practices across regions. These inconsistencies can include mismatched service codes, typographical errors, and incomplete descriptions, all of which complicate the task of categorizing and analyzing the data accurately.

The complexity of the data is a significant challenge in this context. The data set includes hundreds of thousands to millions of invoice entries annually, contributed by various service providers across multiple geographic regions. The large volume and heterogeneity of the data necessitate robust preprocessing and data management strategies to ensure that the input fed into the Large Language Model is of high quality and uniformity.

The other main challenge in this task is managing the classification across an extensive, over 1500, distinct classes. This huge number of categories caused complexity due to the diversity of invoice types, services, and materials. This challenge is associated with the skewed distribution of invoices across these classes, as some classes have many invoices while others are rare. Therefore, it causes the other common issue in the dataset, which is the imbalance nature of the data. Some categories, such as those related to routine maintenance or standard equipment, are overrepresented, while others, particularly specialized services or emergency repairs, are underrepresented. This imbalance can bias the model towards more frequent categories, potentially hindering its performance on less common but critical spending types.

The scalability of data processing systems to handle such large datasets efficiently and the adaptability of these systems to accommodate data from new sources or regions are ongoing challenges. Each region may introduce unique invoicing formats or operational terminologies, requiring continuous adjustments to the data processing and model training frameworks.

Data Cleaning

Data cleaning, an important part of any machine learning process, ensures that the model is trained on high-quality, relevant data and that the model's learning process is not obstructed or misled by extraneous noise. The initial step in our data cleaning process is to identify missing values and for critical features providing appropriate infilling. Any other features with a missing value percentage above 60%, are removed from the training data. Selection of the most relevant features for the model is a critical step. This process involves analyzing the importance of various attributes as well as re-training those that significantly impact the accuracy of the results.

Additionally, the reliability of invoice labels across different regions is also evaluated. Based on this evaluation, regions with unreliable labeling and old regions with no new invoices are excluded from the dataset. This selective exclusion helps maintain the quality and consistency of the training data, avoiding the bias or noise that could degrade the model's performance.

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We also developed a customized dictionary specific to our data that maps acronyms and abbreviations to their full word equivalents. This expansion of shortened forms into full words enhances clarity and uniformity across the text data, facilitating better understanding and processing by the models.

Data Partitioning

In this project, an effective data partitioning strategy is very important for training, validation and testing the model under realistic conditions. This has been done based on the date each invoice was created, ensuring that the model is trained on historical data, validated on recent data, and tested on the most current data to reflect real word operational scenarios. Our data partitioning is designed to maximize learning by using three years of historical data while ensuring robust validation and testing using six months of the most recent data. By regularly updating the partitioning, newer historical data is included in the training process, ensuring that the model has the most up to date information to provide accurate predictions for most recent data.

Model Building and Deployment

Transitioning from data preparation, our next critical phase involves the robust model construction and strategic deployment of our machine learning model. This stage converts clean and structured data into predictive insights that are essential for informed decision-making.

Embedding

In this project focused on optimizing invoice processing, we utilized an advanced transformer model for Language embedding. The model is specifically chosen for its ability to understand and process complicated details within extensive invoice texts. Each invoice is converted into a high-dimensional vector that effectively captures its semantic essence, therefore facilitating more accurate and efficient categorization. The architecture of the transformer is adept at managing the lengthy and detailed invoice descriptions typical in our industry, where each entry might contain multiple lines of detailed service descriptions and contractual terms.

By encoding texts into a high-dimensional vectors, embedding model ensures that each invoice is represented by a dense feature set, enhancing the subsequent classification accuracy by the downstream Llama model.

Model Fine-Tuning

To adjust the Llama (Touvron 2023) model's parameters specifically for the task of invoice categorization, fine-tuning the model is crucial for maintaining its accuracy and adaptability over time. Leveraging Pytorch (Paszke 2019), a flexible deep learning framework, we implemented a process to continuously refine the model based on the latest available data.

The criteria for selecting the model include overall accuracy, macro accuracy, and percentage of correct predictions for spend values. These metrics ensure that the model not only performs well across a broad spectrum of invoice categories, but also enhances its ability to accurately label high value transactions, which is critical due to significant financial implications.

To improve the efficiency of the training process, we leverage GPU computing capabilities. GPUs provide substantial acceleration compared to CPU due to their parallel processing abilities, significantly reducing training time from days to just hours.

Model Results

The fine-tuned and selected model demonstrated a significant capability in achieving an overall accuracy of above 90% and macro accuracy of above 85% across more than 1500 classes, showing its effectiveness in handling a complex and large-scale categorization task.

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The chart in figure 2 provides the plot of accuracy and invoice volumes across different regions for three months. All business units maintain an accuracy rate above 90%, while there are variations in invoice volumes. It highlights the effectiveness of the current solution in place due to the consistent high accuracy results.

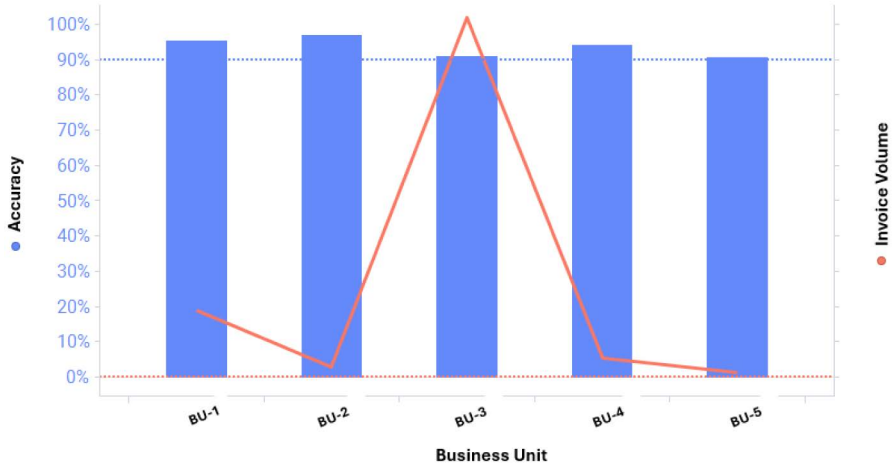

Figure 2—Accuracy and invoice count by business unit (Nov. 2023 to Jan. 2024)

Despite the high overall performance, some classes show lower accuracies compared to others. These variations can be related to several factors:

- Insufficient training samples: Some classes do not have a representative number of samples with which to train the model, resulting in lower accuracies for those classes.
- New classes: When a new class (not seen in the training data) appears in newer data, that class will have lower accuracies until the newer data (containing the class) is included in the scheduled retraining process.

Additionally, the model has consistently achieved high accuracy rates in categorizing invoices with large expenditures. This is critical for the financial management and auditing processes within the Supply Chain. High value invoices often carry more detailed and complex descriptions and the model's success in providing accurate predictions is a great achievement.

Table 1 illustrates a single categorization result from various descriptions using Large Language Models (LLMs). Prior to using LLMs, Supply Chain users had to manually review all available descriptions to categorize each invoice line item. With over 1,500 category options, this task demanded significant manual effort. By incorporating the categorization capabilities of LLMs alongside other organizational parameters such as expenditures across various functions and business units, we can now track costs more accurately in multiple perspectives. This improved model allows for swift and precise categorization of the numerous items purchased, providing a detailed overview of expenditures and identifying potential opportunities for cost savings.

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Tab. 1—Accurate categorization of diverse descriptions by the LLM model

OLA Item Description	Invoice Item Description	Business Unit	Function	...	Category
MEASUREMENT TECH	MEASUREMENT TECH	BU-1	Function-1		MEASUREMENT TECHNICIAN
MEASUREMENT TECH HOUR SERVICE	METER MEASUREMENT AND SAMPLING SVCS. - MEASUREMENT TECH HOUR SERVICE	BU-1	Function-2		MEASUREMENT TECHNICIAN
ALUMINUM INNER BARRELS	CORE ANALYSIS SVC - ALUMINUM INNER BARRELS	BU-2	Function-2		MEASUREMENT TECHNICIAN
CHI.150 PIPE EVAL INTERPRET,PROCESS BY	PIPE EVALUATION INTERPRETATION REPORT - PROCESSING BY TASK	BU-2	Function-3		MEASUREMENT TECHNICIAN
MEASUREMENT TECH HOUR SERVICE	TECH HOUR SERVICE	BU-1	Function-1		MEASUREMENT TECHNICIAN
MEASUREMENT TECHNICIAN I ST	MEASUREMENT TECHNICIAN I ST	BU-1	Function-1		MEASUREMENT TECHNICIAN
(EMPTY)	27P10 PUMP INSPECTION	BU-3	Function-1		MEASUREMENT TECHNICIAN
(EMPTY)	27P10 SWG AND MAXIPROFILE GASKETS	BU-3	Function-1		MEASUREMENT TECHNICIAN
MEASUREMENT TECHNICIAN III ST	WILLIAM G - COMPLETIONS SUPERVISOR/SUPERINTENDENT IV	BU-1	Function-2		MEASUREMENT TECHNICIAN

Further, by aggregating data across various dimensions for each expenditure, we enhance our analytical capabilities, which in turn opens new business opportunities. For instance, we can now monitor inflation and deflation trends in critical commodities like tubular and downhole tools, aiding in long-term strategic planning. Furthermore, an analysis of diesel expenditure in the Permian region highlighted discrepancies in the unit prices from various vendors for the same assets and activities. This finding has significant implications for refining our sourcing strategies in the future. Utilizing our category matrix, we can explore cost-saving opportunities across business units, functions, and suppliers, optimizing our operational efficiency and financial performance.

Model Deployment

The deployment of the fine-tuned Llama model into the operational environment of the oil and gas industry is structured to ensure seamless integration and continuous performance monitoring.

The diagram in figure 3 visualizes the workflows involved in the prediction and re-training processes for our invoice categorization tasks scheduled to start automatically.

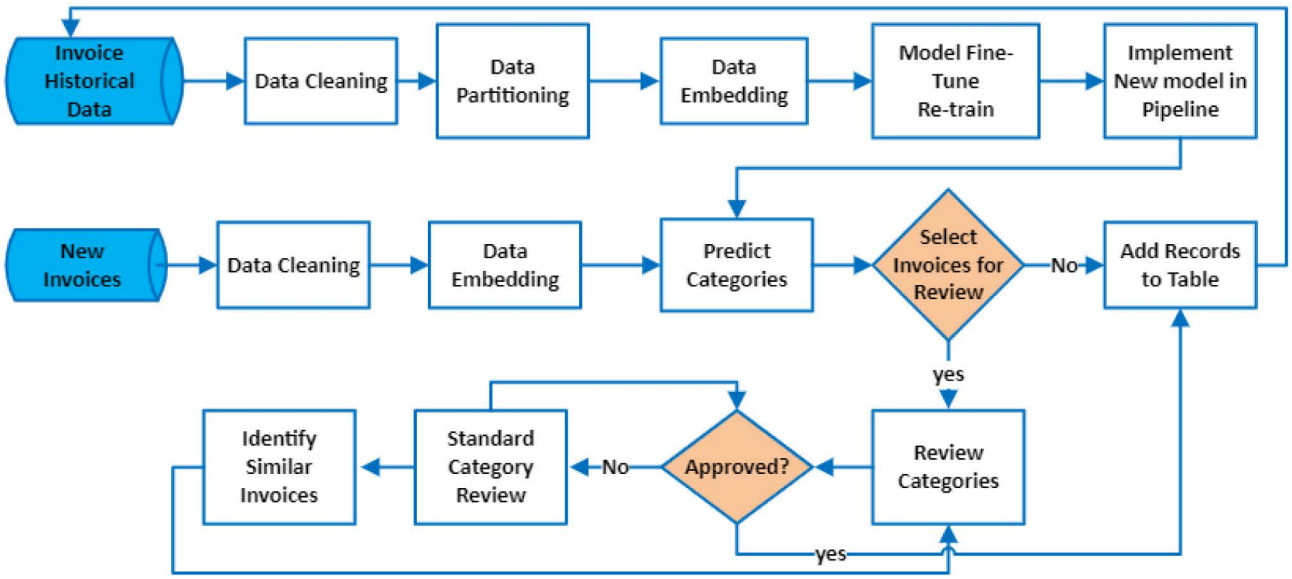

Figure 3—Model deployment workflow diagram

Model Monitoring and Updates

Model monitoring and continuous learning have been implemented to ensure the model maintains high performance and adapts to changes in data patterns and/or business processes.

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Dynamic Updates

A custom dashboard using Streamlit is created to track the model's performance metrics in real-time, providing clear visibility into accuracy rates, processing times, and error rates. The dashboard provides users with invoice predictions as well as corresponding confidence scores. This interface helps end users perform quality control (QC) on model predictions. Supply Chain users can correct any misclassifications directly within the dashboard to enhance the data quality for retraining.

Model Re-Training

The model is scheduled for periodic retraining to adapt to new trends or changes in the invoicing procedures. The comprehensive retraining process is designed to incorporate new data and QC'ed labels, allowing the model to learn from the enriched dataset for improving its accuracy. The retrained model is then redeployed in the operational environment to reflect these improvements.

Conclusions

In this project, we faced the challenge of classifying over a thousand categories for Supply Chain organization in a timely manner. We have implemented sophisticated techniques to manage class imbalance and enhance the discriminative power of the model. By harnessing advanced AI techniques such as embedding, LLM, and leveraging GPU technology for computational efficiency, we have established a robust system capable of handling large-scale data with high accuracy. Our methodology of temporal data partitioning reflects a nuanced understanding of operational dynamics, ensuring that the model is trained on historical data while validated and tested against the most current data available. This approach mitigates potential overfitting and prepares the model for real-world applications by closely simulating the operational environment.

The strategic integration of a user-driven Streamlit dashboard has been instrumental in refining model performance, enabling real-time validation and correction of predictions. This feedback loop not only enhances the model's accuracy but also promotes a collaborative environment where machine learning and human expertise synergize to improve decision-making processes.

This project sets a new benchmark for the application of AI in the oil and gas sector, demonstrating not just the feasibility but also the extensive benefits of AI in streamlining complex operational processes. As we move forward, continuous improvement through model retraining, and updates based on user feedback and new data, will remain central to our approach, ensuring that our solutions remain at the cutting edge of technology and industry needs. Ongoing efforts in innovation using AI are leading to more sophisticated, efficient, and economically viable operational solutions within the industry.

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